Climate discipline — implications and where the work goes next

Status: the synthesis, written down. This is the "so what" of the climate discipline's verified findings, plus the next-work roadmap with an explicit compute-feasibility split. Numbers here are pulled from the guarded modules (deficit, cmip6_drift, ace_fidelity, ecs_model, trust_board, reference_model), never from memory.

The bottom line

Running every model through the strictest checks we can build **does not change the answer — it confirms the mainstream one, and shows why it is right.** The models that actually conserve physics land at ~3 °C sensitivity and ~5 °C of 2100 warming on the current path — the IPCC consensus — reached by trust-gating rather than averaging. That is an independent confirmation. The path there exposed that most of the field cannot even conserve energy, and that is the quantified defect nobody else has.

The implications (five, in order of consequence)

1. Climate "verification" is mostly a category error

"Skillful" (ACE's 93%, every model's skill score) means **agrees with another model, not right about the Earth. 12 of 19 CMIP6 models leak energy** — broken physics, not a tunable bias — yet they are averaged into the projections policy runs on. Oreskes' 30-year-old "you cannot verify a climate model" is now a number: a 1% certificate needs ~18,000 validated observables; we have 4. So "verified climate prediction" is, today, a misnomer — with a number saying exactly how much.

2. Policy: sobering, and it cuts both ways

- **It lets nobody off the hook.** The one model that passes all three gates still projects 4.6–5.2 °C on SSP5-8.5. Verification does not soften the warning; it sharpens it by deleting the untrustworthy models and confirming the trustworthy one.
- **It kills the "hot model" tail honestly.** The scary 5–6 °C ECS models are disproportionately the ones that leak energy. Dropping them on a physics basis (not a convenience basis) lands the ensemble at 2.7–3.1 °C.
- **It hands a decision-maker one sentence:** "The models that conserve physics say ~3 °C sensitivity and ~5 °C on this path; here is exactly why the other 38 are excluded."

3. Spend on measurement, not simulation

The deficit says the bottleneck to a better prediction is **measured data, not compute**. 68% → 1% certificate needs ~18,000 validated observables — that is observational systems (ocean, ice, atmosphere), not a faster supercomputer or

another emulator. Money into "better models" is spending on the wrong wall.

4. AI climate models: a real, urgent caution

ACE's "100× faster" is real, but its "verification" is against a reference GCM — itself a leaky model — never the Earth. The AI-emulator revolution is being graded on the wrong target. Before these replace physics models in policy, they need reality-fidelity checks, not reference-fidelity. The sector is not hearing this yet.

5. The method is the asset, not the numbers

The same spine that found "the future cannot be verified" in string theory found it in climate, and will find it in drugs, economics, AI. The reusable asset is the trust board — per-model trust scores you can point at *any* quantity. The numbers will move; the way we get the trustworthy number is what transfers.

Where the work goes next — with the compute split

Rob's rule: **if we can do the compute, we do it and figure it out; if we cannot, we name the wall and the way around it.** Every next step is tagged with its compute requirement and its verdict.

- **#1 · GISS-E2.1 sea-level by 2100** (finish the board) — compute: fetch the full Nazarenko paper body. Verdict: doable now (no compute, a text fetch).
- **#2 · 2100 warming for the full trusted set** (14 models, not just GISS) — compute: compile per-model SSP warming from CMIP6 archives. Verdict: doable (a fetch + small table, no heavy compute).
- **#3 · Widen conservation to all 53 models** (trust board 17→53 with drift data) — compute: recompute energy/mass/salt drift from raw CMIP6 piControl zarr. Verdict: doable — mesh job (studio-linux; xarray/zarr/dask).
- **#4 · Second-source the drift to all 19** (the documented "natural next step") — compute: raw CMIP6 zarr recompute. Verdict: doable — mesh job, same toolchain as #3.
- **#5 · The citable climate paper** (deficit + drift + ACE + ECS + predictions) — compute: writing + the existing PDF renderer. Verdict: doable now.
- **#6 · The string-landscape h12≥5 sampling** (the physics frontier) — compute: JAXVacua on the mesh (~96 GB / speed problem). Verdict: other discipline, owned by main, not this branch.

The walls, and the way around each (so "cannot" is never a stop)

- **Sea-level (#1):** *wall = unmeasurable-in-hand* — the number is in the paper body, not the abstract. Move: fetch the full text (the Crossref record exposed a full-XML text-mining URL).
- **Conservation widening (#3, #4):** *wall = intractable-on-Mac* — xarray/zarr are not installed here and the data is a streaming fetch. Move: it is a mesh job, not a Mac job — install the toolchain on studio-linux (a build task, per "no limitations on the Rob Mesh") and dispatch.

- **Running GISS-E2.1 or the full CMIP6 ensemble ourselves:** *wall = intractable*

— supercomputer-scale, and not our goal. Move: use the published projections (what we already do); the verification we add is the value, not the simulation.

What changes the answer (honest caveats)

- The strict gate collapses to **one family** (GISS); 2.71 K is one family's view.
- GISS's own ECS (2.9–3.0 °C) differs from the cross-model 2.71 K — method-dependence is real.
- The drift data covers 19 of 53 models; the other 34 are scored on two axes, not three.

None of that reverses the headline: **trust the conserving models, and the answer is ~3 °C / ~5 °C — alarming, defensible, and it says the fix is measurement, not more models.**

Reproduce the numbers

```
PYTHONPATH=src python3 -m pacland.climate.deficit
PYTHONPATH=src python3 -m pacland.climate.trust_board
PYTHONPATH=src python3 -m pacland.climate.ecs_model
PYTHONPATH=src python3 -m pacland.climate.reference_model
```